

# Macroeconomic Theory and Policy: Lecture W6 (Ch.13)

# What are You Going to Learn (Ch. 13)

- The IS and MP curves combine to get the AD curve
- The Phillips curve can be shown as the AS curve
- Movements in the economy can be shown in the AS/AD model
- Modern monetary policy theories
- The appropriate policy response to shocks

# Monetary Policy Rules and Aggregate Demand

Recall that:

- IS Curve:

$$\tilde{Y}_t = \bar{a} - \bar{b}(R_t - \bar{r})$$

- MP curve: Central bank pick  $R_t$  exogenously - so far

- Philips Curve:

$$\Delta\pi_t = \bar{\nu}\tilde{Y}_t + \bar{o}$$

- This model implies high short-run output ( $\tilde{Y}_t$ ) increases inflation ( $\Delta\pi_t > 0$ )
- The central bank faces this trade-off and chooses the interest rate

# Simple Monetary Policy Rule

- A monetary policy rule is a guideline for central bankers under certain economic circumstances, when output or inflation changes
- The following simple rule have the real interest rate a function of the inflation rate:
  - ▶ If inflation is high (low), the interest rate should increase (decrease)

$$R_t - \bar{r} = \bar{m}(\pi_t - \bar{\pi})$$

- The new component is
  - ▶  $\bar{\pi}$ : Inflation target
    - ★ As of now, Canada, the United States, United Kingdom, euro area, and Japan all have an explicit target of 2 percent
    - ★ Treat it as the steady-state rate of inflation
  - ▶  $\bar{m}$ : dictates how aggressively monetary policy responds to inflation

# The AD Curve

- Put together the IS curve and the policy rule (sub in  $R_t - \bar{r}$ )

$$\tilde{Y} = \bar{a} - \bar{b}\bar{m}(\pi_t - \bar{\pi})$$

- Short-run output is a function of the inflation rate
- Invert the equation:

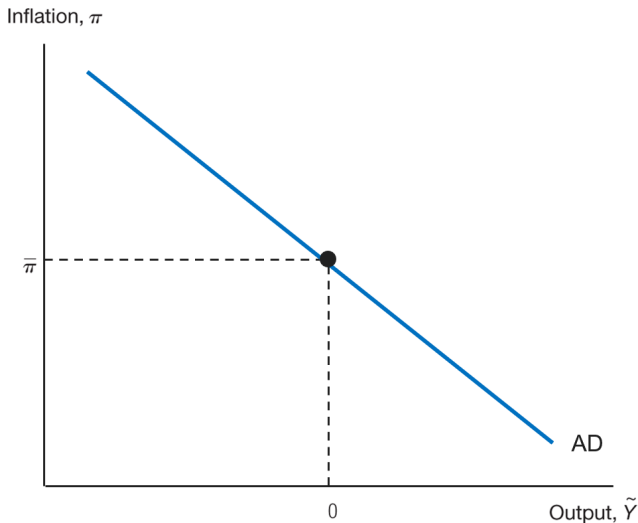
$$\tilde{Y} - \bar{a} = -\bar{b}\bar{m}(\pi_t - \bar{\pi})$$

$$\frac{\tilde{Y} - \bar{a}}{-\bar{b}\bar{m}} = (\pi_t - \bar{\pi})$$

$$\pi_t = \frac{\tilde{Y} - \bar{a}}{-\bar{b}\bar{m}} + \bar{\pi} = \frac{\bar{a}}{\bar{b}\bar{m}} + \bar{\pi} - \frac{1}{\bar{b}\bar{m}}\tilde{Y}$$

# The AD Curve

- If inflation is above (below) its target, the Central Bank raises (falls) rates and output falls (rises) below potential



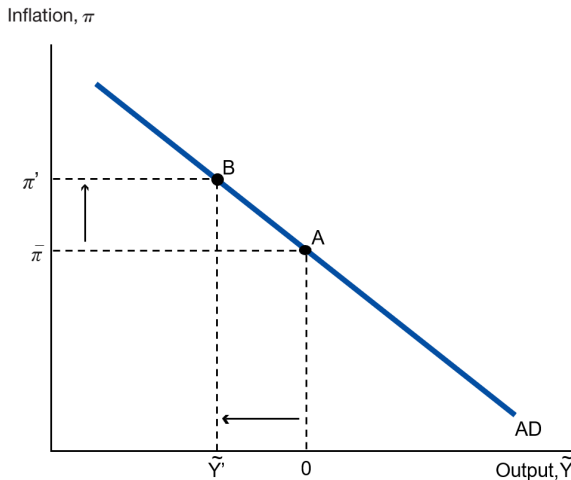
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# The AD Curve

- The AD curve combine the central bank's choice of the interest rate, and, consequently, short-run output, based on the current inflation rate
  - ▶ Change in inflation: a movement along the AD curve
  - ▶ Change in  $\bar{m}$ : change the slope of the AD curve
  - ▶ Changes in the parameter  $\bar{a}$  or  $\bar{\pi}$ : AD curve shifts

# The AD Curve after an Inflation Shock

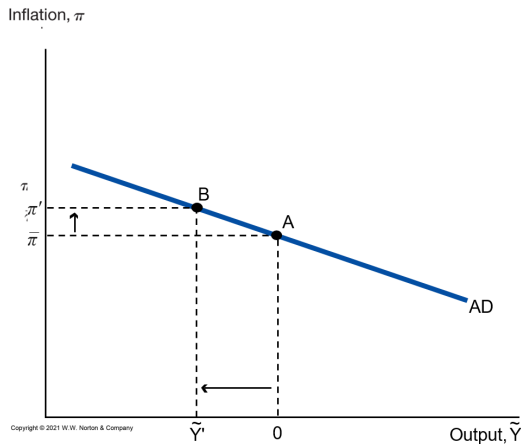
- Seeing an increase in inflation, the monetary policy rule dictates that the central bank increase the interest rate, thereby reducing short-run output



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# An Aggressive Monetary Policy Rule

- A high value of  $\bar{m}$  means the monetary policy rule produces a large recession in order to fight a given increase in the inflation rate
- A one-percentage-point increase in the inflation rate dictates a large decline in short-run output



# The Aggregate Supply Curve

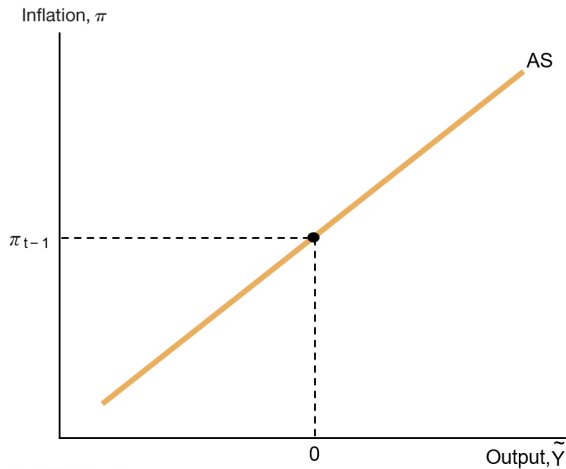
- The Phillips curve relates the current rate of inflation to short-run output
  - ▶ It describe the price-setting equation used by firms in the economy
  - ▶ Also, firms supply good and services

$$\pi_t = \pi_{t-1} + \bar{\nu} \tilde{Y}_t + \bar{o}$$

# The Aggregate Supply Curve

- When there are no inflation shocks ( $\bar{o} = 0$ ) and when output is at potential ( $\tilde{Y} = 0$ ), firms simply expect the current rate of inflation to prevail (i.e.,  $\pi_t = \pi_{t-1}$ )
- The AS curve will shift due to
  - ▶ change in  $\tilde{Y}$ 
    - ★ Higher output today leads firms to raise their prices by more, which increases inflation today and tomorrow
  - ▶ change in the inflation shock parameter
- These shock would carry over to future period until another shock cancel out the effect

# The Aggregate Supply Curve



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# The AS/AD Framework

- Combining the AS and AD curve
- Two equation: AS and AD
- Two unknowns: inflation rate and short-run output

$$\tilde{Y} = \bar{a} - \bar{b}\bar{m}(\pi_t - \bar{\pi})$$

$$\pi_t = \pi_{t-1} + \bar{\nu}\tilde{Y}_t + \bar{o}$$

# The Steady State

- We will first start with steady state, then study the dynamic of the model
- In the steady state
  - ▶ The endogenous variables are constant over time (i.e.,  $\pi_t = \pi_{t-1} = \pi^*$ ,  $\tilde{Y} = 0$ )
  - ▶ No shocks to the economy (i.e.,  $\bar{o} = 0$ ,  $\bar{a} = 0$ )
- Starting with the AS curve (let  $\pi_t = \pi_{t-1} = \pi^*$ ):

$$\bar{\pi} = \bar{\pi} + \bar{\nu} \tilde{Y}_t \implies 0 = \bar{\nu} \tilde{Y}$$

$$\tilde{Y} = 0$$

- The AS curve is consistent with the steady state

# The Steady State

- From the AD curve ( $\tilde{Y} = 0$  and  $\bar{a} = 0$ )

$$\tilde{Y}_t = \bar{a} - \bar{b}\bar{m}(\pi_t - \bar{\pi})$$

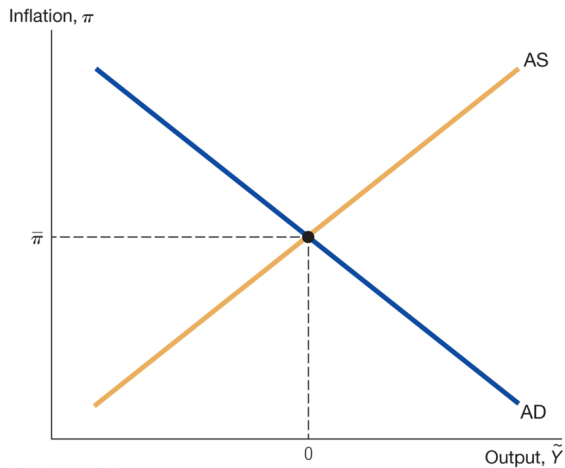
$$0 = 0 - \bar{b}\bar{m}(\pi_t - \bar{\pi})$$

- This implies

$$\pi_t = \pi_{t-1} = \bar{\pi}$$

- AD curve is also consistent with the steady state

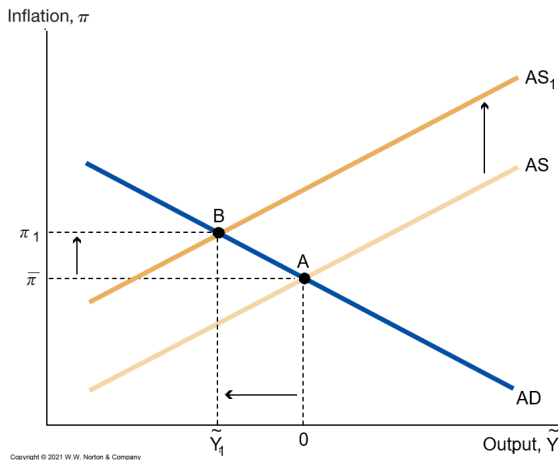
# The AS/AD Framework



# Macroeconomic Events: Event 1—An Inflation Shock

- Unless specific otherwise, the economy begins in steady state
- Suppose there is a increase oil price for one period
  - ▶ The parameter  $\bar{\omega}$  is positive for one period
  - ▶ The inflation rises
  - ▶ The AS curve will shift up as a result
- The monetary policy rule states that the increase in inflation be fought by an increase in the real interest rate, reducing short-run output

# The Initial Response to an Inflation Shock



- Stagflation: Combination of a recession and inflation
- But, that is just the initial impact

# Event 1: An Inflation Shock

- In period 1, the inflation would be

$$\pi_1 = \bar{\pi} + \bar{\nu} \tilde{Y}_1 + \bar{o}$$

- In period 2,  $\bar{o}$  return to zero, the AS curve is now:

$$\pi_2 = \pi_1 + \bar{\nu} \tilde{Y}_2 = \bar{\pi} + \bar{\nu} \tilde{Y}_1 + \bar{o} + \bar{\nu} \tilde{Y}_2$$

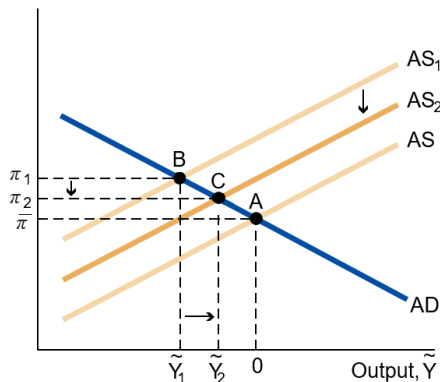
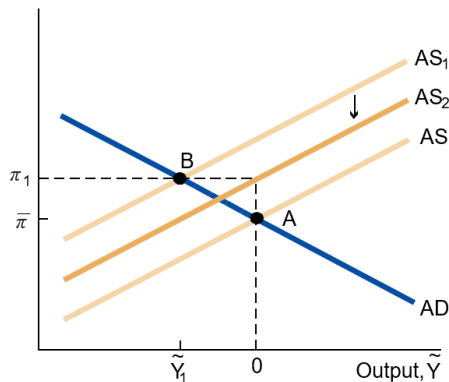
- Since  $\pi_1 > \bar{\pi}$ , the AS curve is not back to steady state
  - ▶ Note the term  $\tilde{Y}_1$  and  $\tilde{Y}_2$  are negative
  - ▶ This implies the inflation would get lower and lower over time!!

# Event 1: An Inflation Shock

- In word,
  - ▶ High inflation from the oil shock
  - ▶ It rises the expected inflation
  - ▶ AS curve does not adjust back to steady state right away
  - ▶ As seen next, inflation slowly falls and model return to its steady state

## Two Periods after an Inflation Shock

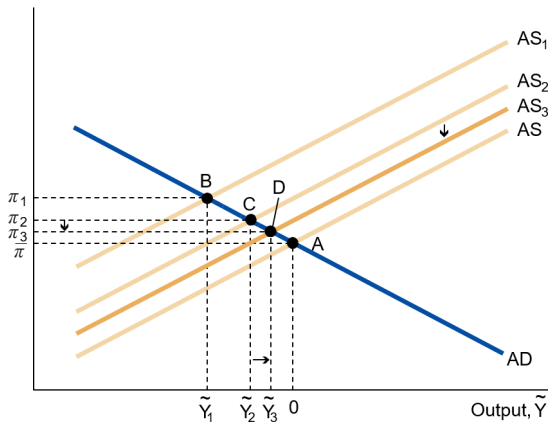
- Because of sticky prices, high inflation expectations adjust only slowly
- At  $\tilde{Y} = 0$ , the AS curve would shift down a bit from  $AS_1$



## Three Periods after an Inflation Shock

- In period three, expected inflation falls further, shifting the AS curve closer to its original position
- Eventually, the AS curve would move back to the steady state position

Three Periods after an Inflation Shock  
Inflation,  $\pi$

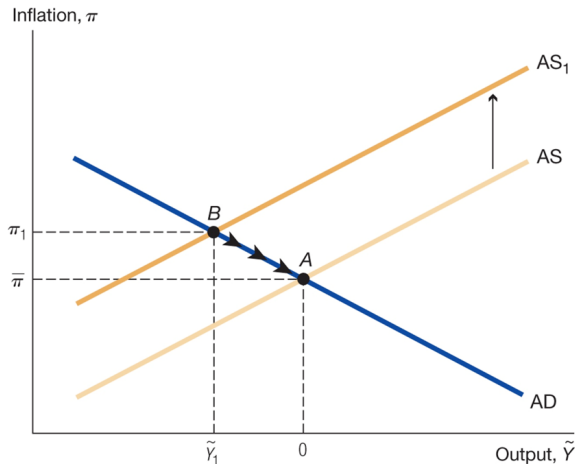


## Event 1: An Inflation Shock

- What we have just saw is similar to the principle of transition dynamics, which we encountered in the long-run model
- Another note is that the movement back to the steady state is fastest when the economy is furthest from its steady state
- In summary, a price shock
  - ▶ raises inflation directly
  - ▶ the central bank induces a recession to fight inflation
  - ▶ keeps inflation higher for a longer period of time due to sticky inflation
  - ▶ results in a prolonged slump
  - ▶ causes the economy to suffer from stagflation

# The Effects of an Inflation Shock

- As expected inflation declines gradually, causing the AS curve to shift back toward its original position
- Sticky inflation assumption is at the heart of this gradual adjustment



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## Event 2: Disinflation

- Suppose policymakers decide to lower the inflation target.

- ▶ The AD curve shifts down, as  $\bar{\pi}$  falls

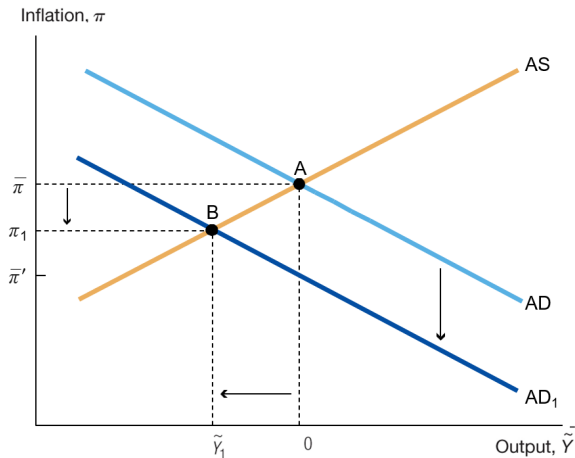
$$\tilde{Y}_t = \bar{a} - \bar{b}\bar{m}(\pi_t - \bar{\pi})$$

- ▶ The new rule calls for an increase in interest rates

$$R_t - \bar{r} = \bar{m}(\pi_t - \bar{\pi})$$

# The Initial Response to Disinflation

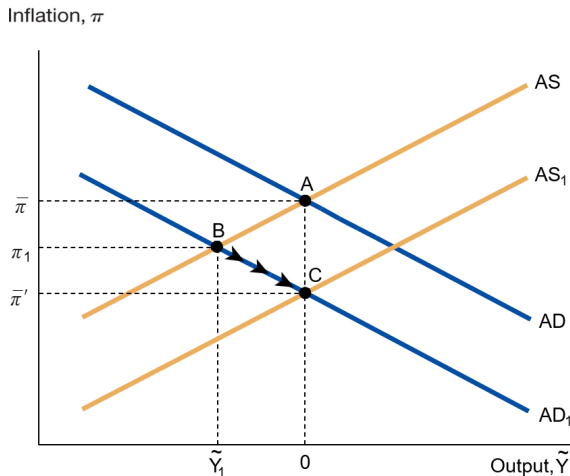
- The reduction in the inflation target shifts the AD curve down



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# The Dynamics of Disinflation

- We can apply the principle of transition dynamics
- Here, the change in the monetary policy rule means that the steady-state rate of inflation has now changed



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## Event 2: Disinflation

- If the classical dichotomy holds in the short run, the AD and AS curves would reach the new steady state immediately
- But there is a sticky inflation. A recession is needed to adjust expectations down
- In the process of changing a monetary policy rule, the credibility of policymakers and the way price setters form their expectations play critical roles
  - ▶ Michelle Alexopoulos (at U of T) and her co-authors analyze effect U.S. Federal Reserve Chairs's word, voice and facial expression when deliver congressional testimonies on financial markets (JME 2024)

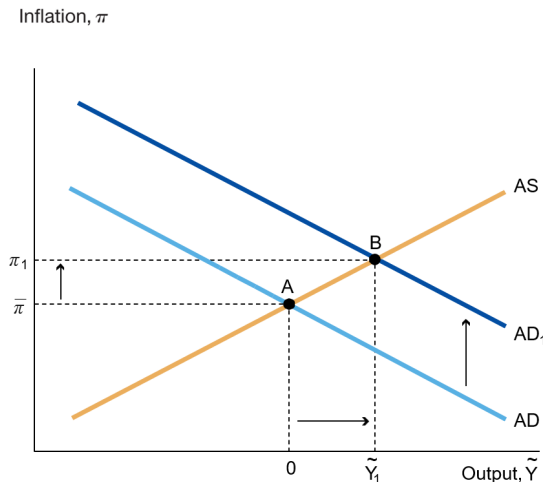
## Event 2: Disinflation

- Summary:

- ▶ The policy change lead to a downward shift of the AD curve
- ▶ The change in the rate of inflation causes the AS curve to shift in the following period
  - ★ Firms adjust their expectation for inflation to account for the new lower inflation rate
- ▶ The inflation rate is still above the target, the central bank keeps actual output below potential
- ▶ Eventually, as inflation rate falls further, the economy will rest in its new steady state

## Event 3: A Positive AD Shock

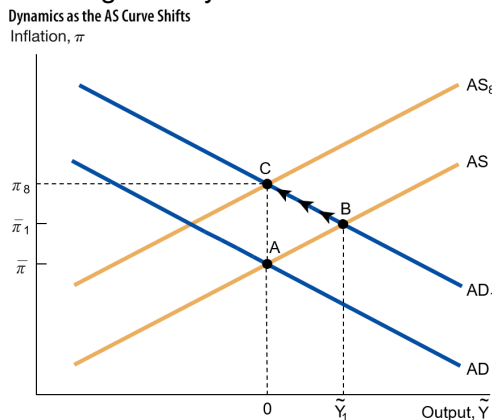
- Suppose there is a temporary increase in the aggregate demand parameter,  $\bar{a}$
- The AD curve will shift out
  - ▶ Increased demand lead to an increase in prices



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## Event 3: A Positive AD Shock

- Following the increase in demand (i.e.,  $\bar{a}$  increases), firms raise prices.
  - ▶ Inflation increases
- As the inflation rise, AS curve shift up
- Short-run output return to zero gradually

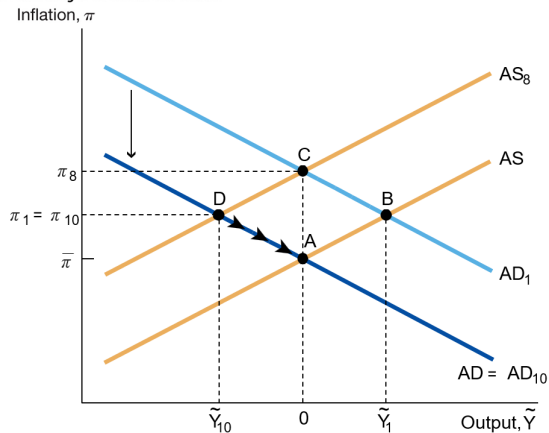


# The Unraveling after the AD Shock Ends

- Recall, aggregate demand shocks are temporary. In the long run,  $\bar{a} = 0$
- When the shock ends, the AD curve shifts back to its original position
  - ▶ This is much like a negative shock because short-run output was already at 0
- The shift back to its original position puts downward pressure on inflation:
  - ▶ Transition dynamics returns the economy to steady state as the AS curve shifts

# The Unraveling after the AD Shock Ends

The Unraveling after the AD Shock Ends



## Event 3: A Positive AD Shock

- The AD shock implies that booms are matched by recessions
  - ▶ The economy benefits from a boom but inflation rises
  - ▶ The way to reduce inflation is by a recession
- The costs of inflation: The economy would have been better off at the original steady state

# Empirical Evidence: Predicting the Central Bank Policy Rate

- Does simple monetary policy rule provides an accurate picture of the federal funds rate?
- Recall the policy rule and the Fisher's equation

$$R_t - \bar{r} = \bar{m}(\pi_t - \bar{\pi})$$

$$i_t = R_t + \pi_t$$

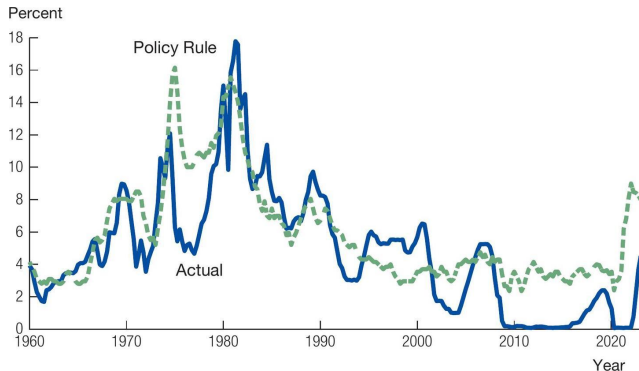
- Use the Fisher equation to write the monetary policy rule in terms of the nominal interest rate:

$$i_t = \bar{r} + \pi_t + \bar{m}(\pi_t - \bar{\pi})$$

- It is commonly known as the Taylor's rule
  - ▶ The Taylor's rule typically also have output build into it

# The Fed Funds Rate, Actual and Predicted

- Predicted level for the fed funds rate using the simple policy rule
  - ▶  $\bar{m} = 1/2, \bar{r} = 2\%, \bar{\pi} = 2\%$
- Some economists infer that monetary policy was excessively loose during 70s, resulting the high inflation in the 70s
- The Fed also departs from the rule during recessions in the early 1990s, in the early 2000s, and since 2008



Source: The FRED database.

# Next Lecture

- Next Lecture will cover Ch.18